

Day 2 — Axial Tilt and the Seasons

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: demonstrate with torch+globe why we have summer and winter.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Uttarāyaṇa* (IAST: Uttarāyaṇa; lit. “northward solar course (Dec 21 – June 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Torch + globe demo.** Hold a globe with the axis tilted $\sim 23.5^\circ$ (mark the North Pole with tape). Shine a strong torch from one direction (the Sun). Slowly rotate the globe and observe which hemispheres get more direct light.
2. **Key observation.** When the Northern Hemisphere tilts TOWARD the Sun (June solstice), it gets more direct sunlight, longer days. Southern Hemisphere experiences the opposite (winter).
3. **Source.** Sūrya-siddhānta and Āryabhaṭīya both note Earth’s globe-shape (*gola*) and the tilt. Āryabhaṭa correctly identifies the rotation cause of day-night ~ 499 CE.
4. **Modern bridge.** NCERT Physics XI ch. 8 gives the modern derivation: Earth’s axial tilt + orbital geometry $\rightarrow$ annual solar declination $\rightarrow$ seasons.
5. **Discussion.** “*Why are January and July temperatures so different — even though Earth is at almost the same distance from the Sun?*” (Answer: tilt, not distance.)

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Axial Tilt and the Seasons.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).